

Hishaam Ali Abbasi

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PROFILE

First Class Robotics Engineering graduate from Queen Mary University of London with hands-on experience in mechanical design, embedded programming, control systems and robotics integration. Designed and tested physical prototypes, diagnosed faults on robotic hardware, and led a workstream within a multidisciplinary team.

EDUCATION

BEng Robotics Engineering First Class Honours

Queen Mary University of London | **IMEchE Accredited** | 2023-2026

Selected modules: Advanced Robotics Systems (90.8%) | Numerical Methods & Data Science (86%) | Control System Analysis & Design (78.9%) | Integrated Design Project (78.8%) | Instrumentation & Measurements (76.1%) | Modelling & Control of Robotic Systems (75.2%)

Sutton Grammar School A Levels: Mathematics (A), Physics (A), Computer Science (B). 12 GCSEs including Maths (9), Computer Science (8) and Physics (8).

EXPERIENCE

Engineering Design and Research Intern

Alvis Robotics, Birmingham | EOD Systems | August 2025 (2 weeks)

- Diagnosed a sensor and camera activation fault on a Northrop Grumman EOD unmanned ground vehicle. Replaced incompatible driver boards in the lighting and camera subsystem to restore operation.
- Diagnosed and resolved a seized active-pitch mechanism on a tracked EOD platform; disassembled the crawler drive and cleared trapped mud to return the unit to operation.
- Supported R&D planning with the Managing Director on next-generation EOD robot development.

Additional experience: Clinical Data Administrator, AT Medics (2021); COVID Clinic Front of House, AT Medics (2020-2021).

PROJECTS

Fracture Immobilisation Trainer - Final Year Dissertation

Blender | ESP32 | Haptics | Fluidics | Sensors | 3D printing | 2025-2026

- Led the anatomical core workstream in a five-person team; designed and manufactured a 1:1 adult-scale forearm chassis in Blender with internal routing for fluidics, sensor wiring and haptic actuators.
- Compared four FDM materials through a cantilever deflection study under progressive loading. Selected PETG-CF based on its measured elastic stability across the test range.
- Designed and 3D-printed moulds and cast layered silicone soft tissue, using Dragon Skin 30A for muscle and Ecoflex 00-30 for skin.
- Integrated Hall-effect sensing at the fracture site with an ESP32 to provide real-time fracture-alignment feedback.
- Contributed to a functional medical training prototype costing under £500.

TurtleBot3 SLAM and Autonomous Navigation

ROS | SLAM | Gazebo | Python | Raspberry Pi | University coursework

- Implemented differential-drive kinematics and ROS-based SLAM using Gmapping, combining LiDAR scans and wheel odometry to build a real-time 2D occupancy map on a physical TurtleBot3.
- Developed a Python obstacle-avoidance node using LiDAR distance thresholds to trigger reactive turns; transferred the navigation stack from Gazebo simulation to physical hardware.
- Resolved ROS/ROS2 package conflicts, operating-system dependencies and networking issues. Configured Ubuntu on Raspberry Pi and a dedicated private network for ROS communication.

PROJECTS - CONTINUED

Formula Student - Suspension System Redesign

Queen Mary Racing | CAD | Ansys FEA | Vehicle dynamics | Oct 2023-Jun 2026

- Redesigned eight suspension components, including wishbones, uprights and bell crank, for a new vehicle platform.
- Used Ansys FEA and geometry optimisation to reduce unsprung mass by 12% while maintaining peak structural stiffness.
- Coordinated component design with powertrain, aerodynamics and chassis teams to meet shared vehicle requirements.

Home Server and Self-Hosted Infrastructure

Linux | Docker | Proxmox | Cloudflare | AWS | Personal project | 2025-Present

- Built a Proxmox VE server on a Lenovo ThinkStation with GPU passthrough to an LXC container for hardware video transcoding; resolved kernel, device-passthrough and driver conflicts.
- Deployed services through Docker Compose, including Nextcloud with 8 TB of storage for four users and encrypted nightly backups to AWS S3 Glacier Deep Archive.
- Configured Cloudflare Tunnel and Zero Trust Access across five custom domains and migrated four websites from Vercel to self-hosted Nginx.

Active Camera Gimbal

C++ | PID control | Embedded systems | Personal project

- Built a two-axis gyroscopically stabilised camera gimbal with custom C++ firmware and real-time PID control loops on an embedded microcontroller, integrating mechanical hardware, electronics and software.

Additional projects: PETG-CF sim racing pedals with load-cell braking; FDM printer upgrades using Klipper, linear rails and remote monitoring. Further case studies at hishaamabbasi.co.uk.

LEADERSHIP

Explorer Scout Leader

Scouting Movement (UK)

- Founded a new Explorer section; plan and deliver weekly sessions, expeditions and residential camps for a group of 18+ young people. Achieved the Chief Scout Diamond Award.

Flight Sergeant

Combined Cadet Force | Sep 2018-Jul 2023

- Taught the CCF syllabus, led teams on residential camps and mentored younger cadets.

TECHNICAL SKILLS

Programming: Python (Advanced) | Java (Advanced) | C++ (Intermediate) | C (Basic) | SQL (Basic) | HTML/CSS/JavaScript (Basic)

Robotics & Control: ROS / ROS2 | SLAM (Gmapping) | PID control | Kinematics | Gazebo | Haptics | Sensor fusion | ESP32 | Arduino | Raspberry Pi

CAD & Manufacturing: Onshape (Advanced) | SolidWorks, Creo, Inventor / Fusion 360 (Intermediate) | Ansys FEA (Basic) | AutoCAD (Basic) | FDM 3D printing and Orca Slicer (Advanced) | Klipper | Material testing | Silicone casting

Infrastructure: Linux (Debian / Ubuntu) | Proxmox VE | Docker / Docker Compose | LXC | Nginx | Cloudflare Tunnel / Zero Trust | AWS S3 / Glacier | Git

Other: Full UK driving licence | Native English | Conversational Urdu